PAPER PRESENTATION

Goal-Oriented Intelligence Path Planning for LLMs Based Drone in Adversarial Conditions

Presented by

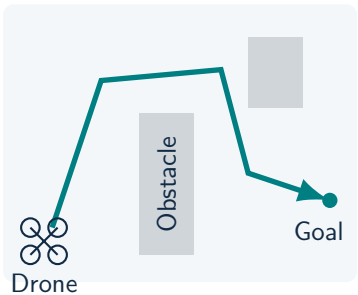
Aarush Ram Anandh

Paper by Muhammad Sohail Farooq et al.

1 Why use language models for drone navigation?

Paper: Goal-Oriented Intelligence Path Planning for LLMs Based Drone in Adversarial Conditions

- Reach a destination safely in unfamiliar surroundings.
- Adapt when obstacles or conditions change.
- Use visual context and instructions to propose routes.



The question for this talk

What useful role can a language model play in a drone navigation system?

Reading the evidence. GPS-denied navigation is a motivation; the reported evaluation is simulated obstacle navigation.

Source: supplied paper, pp. 1, 4–5, Sections I, III–IV.

2 Background: perception, language, and planning

Component	What it does	Role in the paper
YOLOv8	Detects objects in images	Supplies visible obstacle information
LLM	Models token sequences	Interprets instructions; generates structured outputs
VLM	Processes images with text	Proposes routes from maps and instructions
RRT	Searches free space	Generates example waypoint routes
Flight controller	Tracks a trajectory	Executes the processed route

Named models: Gemini 2, ChatGPT-4, Claude 3.5, Grok 3.
No separate model results or clear identity for the final reported numbers.

Reading the evidence. An image detection is not a 3D obstacle position: depth, calibration, and coordinate transforms are also needed.

Source: supplied paper, pp. 4–5, Sections III–IV; paper spells Grok as "Gork" in Section III.

3 Philosophy and operating constraints

Division of labour

Learned perception and planning + conventional trajectory processing and flight control.

Intended roles

1. Perception: identify surroundings.
2. Planning: propose waypoints.
3. Execution: follow a trajectory.

Examples teach output patterns; visual context describes the current scene.

Operating assumptions

- User supplies start and goal positions.
- Observations support obstacle geometry.
- Every route segment needs clearance.
- Moving obstacles need timely replanning.
- Remote inference needs connectivity and low delay.

Reading the evidence. Learned predictions need geometric validation. Reliable replanning frequency and end-to-end latency are not established.

Source: supplied paper, pp. 4–5, Sections III–IV.

4 Architecture: from observations to flight

Optional — if time permits

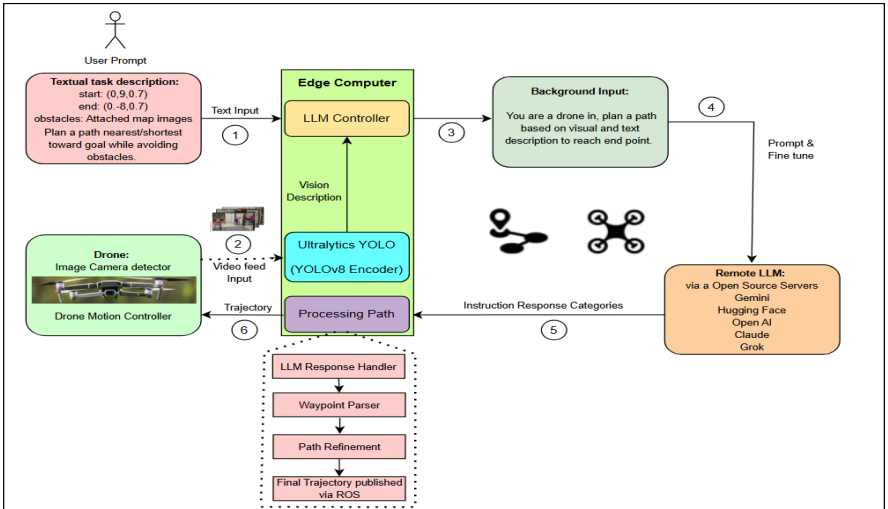


Fig. 3. The methodology proposed by using LLM

Source: supplied paper, p. 4, Figure 3 (original figure) and Section III.

Read Figure 3 in order

1. Prompt + camera
2. YOLOv8 context
3. Remote model
4. Parse + refine
5. ROS + controller

The exact LLM/VLM arrangement is underspecified.

4 Architecture: from observations to flight

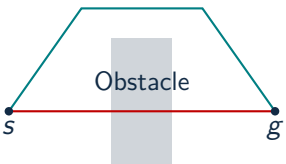
Optional — if time permits

A. Define the planning task

Mathematical reconstruction

$$W = (w_0, \ldots, w_K), \qquad w_0 = s, \quad w_K = g, \qquad L(W) = \sum_{i=0}^{K-1} \|w_{i+1} - w_i\|_2.$$

Validate entire segments



A short path matters only if it is safe.

$$\text{dist}([w_i, w_{i+1}], \mathcal{O}) \geq r + \delta$$

r: drone radius; δ: extra margin.

Safe endpoints can enclose an unsafe segment.

Reading the evidence. The paper does not explicitly optimize this route-length objective or specify the collision-checking algorithm.

Source: supplied paper, pp. 4–5, reconstructed from the navigation task.

4 Architecture: from observations to flight

Optional — if time permits

B. Explain the learned planner

Mathematical reconstruction

$$C = (s, g, V, I), \qquad p_\theta(W \mid C)$$

V: visual context; I: instructions; θ: learned parameters.

Learning a target route encoded as tokens

$$\mathcal{L}_{\mathrm{CE}} = -\sum_j \log p_\theta(y_j^* \mid y_{<j}^*, C).$$

Assign higher probability to the target output sequence.

- RRT-generated waypoints are the apparent supervision source.
- Three visual examples are reported for few-shot prompting.
- Fine-tuning: 105 epochs; training loss ≈ 0.15 near epoch 65.

Reading the evidence. Matching target tokens does not directly optimize safety or length. Prompting and fine-tuning are described, but their exact relationship is unclear.

Source: supplied paper, pp. 4–5, Sections III–IV; loss equation is an explanatory reconstruction.

4 Architecture: from observations to flight

Optional — if time permits

C. Process the proposed route

Interpolation reconstruction

$$q(\alpha) = (1-\alpha)w_i + \alpha w_{i+1}, \qquad 0 \le \alpha \le 1.$$

Insert intermediate positions along each segment



Reported pipeline: response handler → waypoint parser → path refinement → trajectory via ROS → motion controller.

- Reported interpolation spacing: $0.1\,\mathrm{m}$.
- A geometric route still needs timing and feasible control commands.

Reading the evidence. More points do not remove collisions or guarantee physically feasible turns. The paper does not specify the controller or timing law.

Source: supplied paper, p. 4, Figure 3 and Section III.

5 Results, limitations, and our extensions

Reported simulation results

Environment	Success	Time
Simple	95%	65.7 s
Moderate	92%	65.4 s
Complex	91.5%	69.2 s
Dynamic	89%	72.0 s

Figure 4 also calls these percentages "efficiency"; no separate efficiency definition is clear.

Clearance conflicts: 2 m vs. 1.2 m.
RL is an author-proposed future direction.

Limitation $\longrightarrow$ our proposed extension

- No controlled LLM comparison
  Compare conventional, LLM, and hybrid planners on identical maps.
- No demonstrated route safety guarantee
  Validate every segment; use a conventional fallback.
- Missing trials, model identity, and timing
  Report configuration, repeated trials, uncertainty, and planning latency.

Takeaway

A plausible integration; the experiments do not establish the LLM's added value beyond conventional planning.

Source: supplied paper, pp. 4–6, Sections III–V, Figure 4 and Table I; extensions are ours.